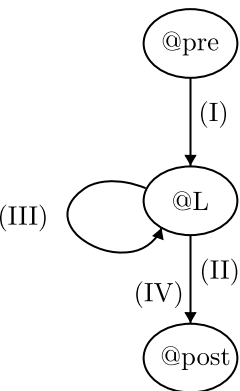


@pre: $0 \leq l \wedge u < |a|$

@post: $rv = \text{true} \iff \exists i: l \leq i \leq u \wedge a[i] = e$

```
Bool LinearSearch(int[] a, int l, int u, int e)
{
    @L:  $l \leq i \wedge (\forall j: l \leq j < i \implies a[j] \neq e)$ 
    for(int i := l; i ≤ u; i := i+1){
        if(a[i]=e)
            return true;
    }
    return false;
}
```



(I)@pre
i:=1;
@L

(III)@L
assume $i \leq u$;
assume $a[i] \neq e$;
i:=i+1;
@L

(II)@L
assume $i \leq u$;
assume $a[i] = e$;
rv:= true;
@post

(IV)@L
assume $i > u$;
rv := false;
@post